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Paleomagnetism, rock magnetism, and fission track dating of the Kurihashi Granodiorite in northeast Japan: Evidence for a Cretaceous hydrothermal event and differential rotation along the eastern Eurasian margin

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Abstract. Samples from the Late Cretaceous Kurihashi Granodiorite on the Pacific coast of northeast Japan are divided into two groups based on paleomagnetic data. One group (A) preserves stable single thermoremanent magnetizations (TRM) with large westerly deflected declinations. The other group (B) has two or three distinct components of natural remanent magnetization (NRM): components with high, middle, and low unblocking temperatures show westerly declinations (nearly identical with the stable TRM directions of group A), as well as northerly and easterly declinations. Rock magnetic experiments indicate that the dominant magnetic mineral of both groups is Ti-poor titanomagnetite. Group A samples contain superparamagnetic (SP) titanomagnetite because they show a frequency dependence with the initial susceptibility measurements: Hysteresis parameters of group B samples are consistent with a mixture of single-domain (SD) and multidomain (MD) grains. Electron microscopy confirms the presence of MD size magnetite in the group B samples. Fission track thermochronology indicates that the Kurihashi Granodiorite underwent a post cooling hydrothermal event during the Late Cretaceous; thus the northerly and easterly NRM components seem to have a thermoviscous origin. Many samples have anomalously high unblocking temperatures (up to 560°C) compared to predictions of SD theory and probably result from long thermal demagnetization "tails" of thermoviscous remanent magnetizations (TVRM) carried by MD magnetite. A strong TVRM was acquired during the hydrothermal event and is correlated with the long normal-polarity superchron of the geomagnetic field. Because highly oblique (left-lateral) subduction was active along the eastern Eurasian margin during the same period, the primary TRM and TVRM components record counterclockwise rotation (~120°) of the Kurihashi Granodiorite and remarkable wrench deformation along the continental margin.

1. Introduction

Honshu Island is part of the island arc chain along the eastern Eurasian margin. Recent paleomagnetic studies [e.g., *Otofuji et al.*, 1985a, b; *Tosha and Hamano*, 1988; *Otofuji et al.*, 1991] have determined that the island was rifted and then drifted from the continent during the Miocene opening of the Japan Sea (Figure 1). Although most of these studies regard Honshu Island, explicitly or implicitly, as having evolved from a number of accretionary complexes and volcanic arcs along the continental margin since Mesozoic time, the tectonic coherency of the island's northern half (northeast Japan) is unclear considering the large scatter in the available paleomagnetic data. Northeast Japan consists of Paleozoic and Mesozoic rocks intruded by voluminous Cretaceous granites, overlain by thick Neogene volcaniclastics and marine sediments. Cretaceous to early Miocene paleomagnetic directions for northeast Japan are generally characterized by large westerly deflections, which have been attributed to a coherent counterclockwise rotation during the fan-shaped, back arc opening of the northern Japan Sea [*Otofuji et al.*, 1994]. However, *Otofuji et al.* [1997] reported extremely shallow inclinations from Cretaceous and Paleogene welded tuffs on the Pacific coast (star 1 in Figure 1a) and argued for Cenozoic northward translation of a part of northeast Japan using Pacific plate motion. Their theory, however, must be supported by a plausible mechanism which can generate voluminous acidic igneous rocks among oceanic plates. On the other hand, significant easterly deflection was reported from Paleogene welded tuffs on the Japan Sea coast (star 2 in Figure 1a) [*Otofuji et al.*, 1985b] and was attributed to differential rotation near the Nihonkoku-Miomote Tectonic Line. As this tectonic line is one of many NNW-SSE trending faults in northeast Japan (Figure 1a [*Oide*

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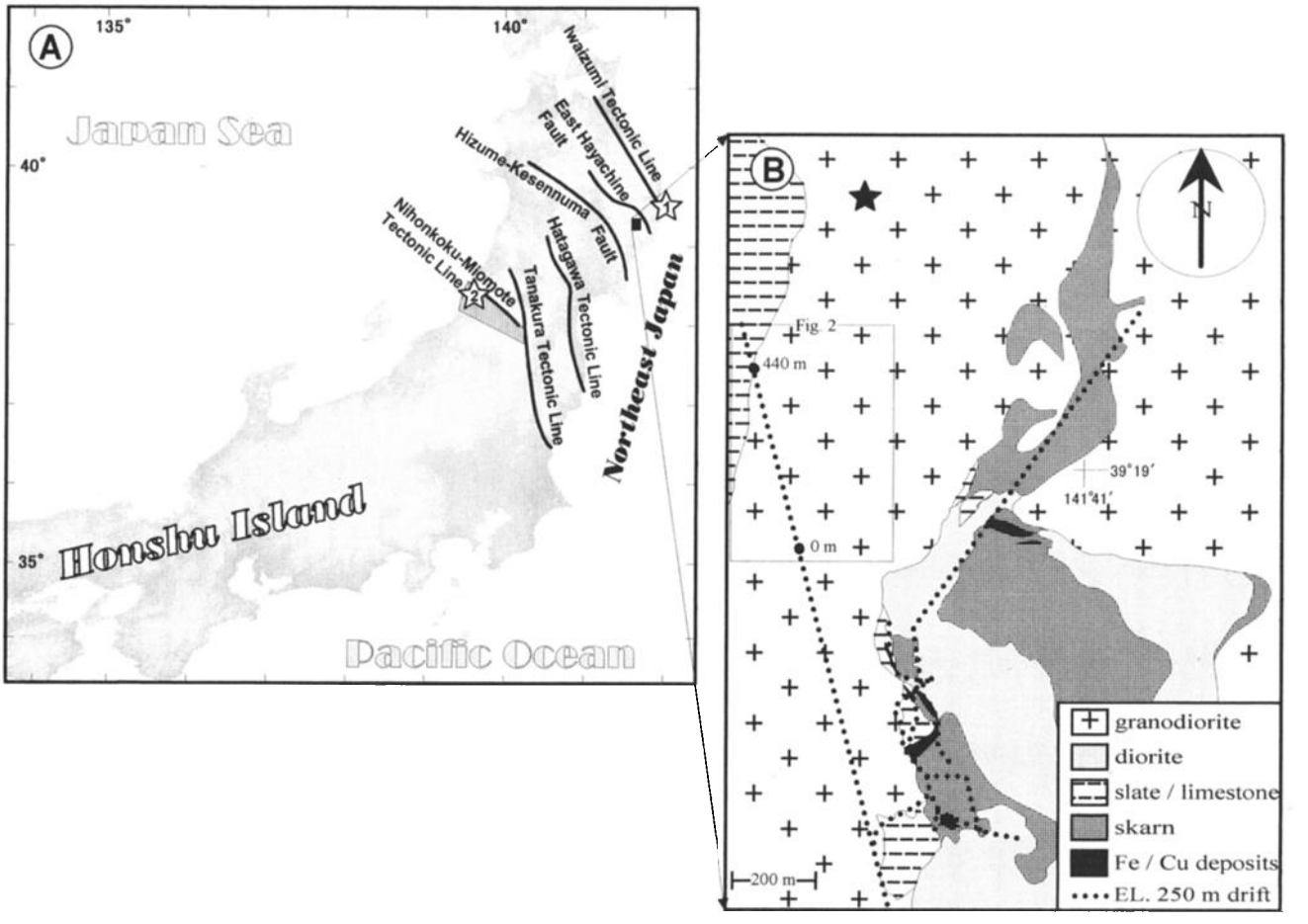


Figure 1. (a) Regional index map showing large NNW trending faults in northeast Japan [*Oide et al.*, 1989]. Stars 1 and 2 in the index map denote paleomagnetic data presented by *Otofuji et al.* [1997] and *Otofuji et al.* [1985b], respectively. Hatched zone adjacent to star 2 denotes dextral deformation zone by *Yamaji et al.* [1999]. (b) A geologic map of the study area [*Itoh et al.*, 1998a], which is drawn at 250 m above sea level (masl) (EL. on figure). Solid star denotes projected sampling locality of fluid inclusion analysis (300 m above the mapped level) by *Amano et al.* [1998].

et al., 1989]), such rotation implies the rearrangement of terranes by transcurrent motions along these faults.

Determination of the timing and extent of rotational events based on reliable paleomagnetic data and radiometric ages is essential to clarify the tectonic history of the Eurasian plate margin. In this article, we present a paleomagnetic and chronologic study of the Kurihashi Granodiorite near the Pacific coast of northeast Japan. A preliminary investigation [*Itoh et al.*, 1998a] showed that the plutonic rock can be divided into two groups based on paleomagnetic data: one sample group (A) preserves a stable single component of thermoremanent magnetization (TRM), and the other sample group (B) has two or three distinct components of remanence. The mean characteristic direction of the TRMs of group A has a large westerly deflection (Figure 2) consistent with other directions reported from Cretaceous plutonic rocks in northeast Japan [*Ito and Tokieda*, 1986]. It is expected therefore that analysis of the multicomponent remanent magnetization will indicate the rotational process in our study area. In addition, fission track thermochronology will provide us with the timing of remanence acquisition events, namely, the initial cooling of the plutonic body and later thermal event that is anticipated from fracture-filling mineral assemblages [*Amano et al.*, 1998]. Through this integrated study, we aim to clarify the deformational process dominant along the continental margin since the Cretaceous.

2. Geological Setting and Sampling

Large bodies of Cretaceous plutonic rocks crop out on the Pacific side of northeast Japan. Most radiometric ages for them range from 120 to 110 Ma [e.g., *Kawano and Ueda*, 1965]. The Kurihashi Granodiorite near the East Hayachine Fault (Figure 1a) intrudes Paleozoic and Mesozoic sediments and has been intensively studied because of its economic iron and copper mineralization [*Oide et al.*, 1989]. Although Cretaceous plutonic rocks in northeast Japan in most surface outcrops are leucocratic and show unstable rock magnetic nature [*Okuma and Kanaya*, 1990; *Okuma*, 1993], subsurface samples are characterized by higher stability in remanent magnetization [*Itoh et al.*, 1998a]. In this study, samples were collected in a drift at ~600 m below the surface in the former Kamaishi Mine, which was maintained as an in situ test site for geological disposal of high-level radioactive waste. Voluminous studies on structural geology, mineralogy, and radiometric ages [*Yoshida et al.*, 2000] helped us to determine the sampling scheme. A horizontal geologic section at 250 m above sea level (Figure 1b) shows

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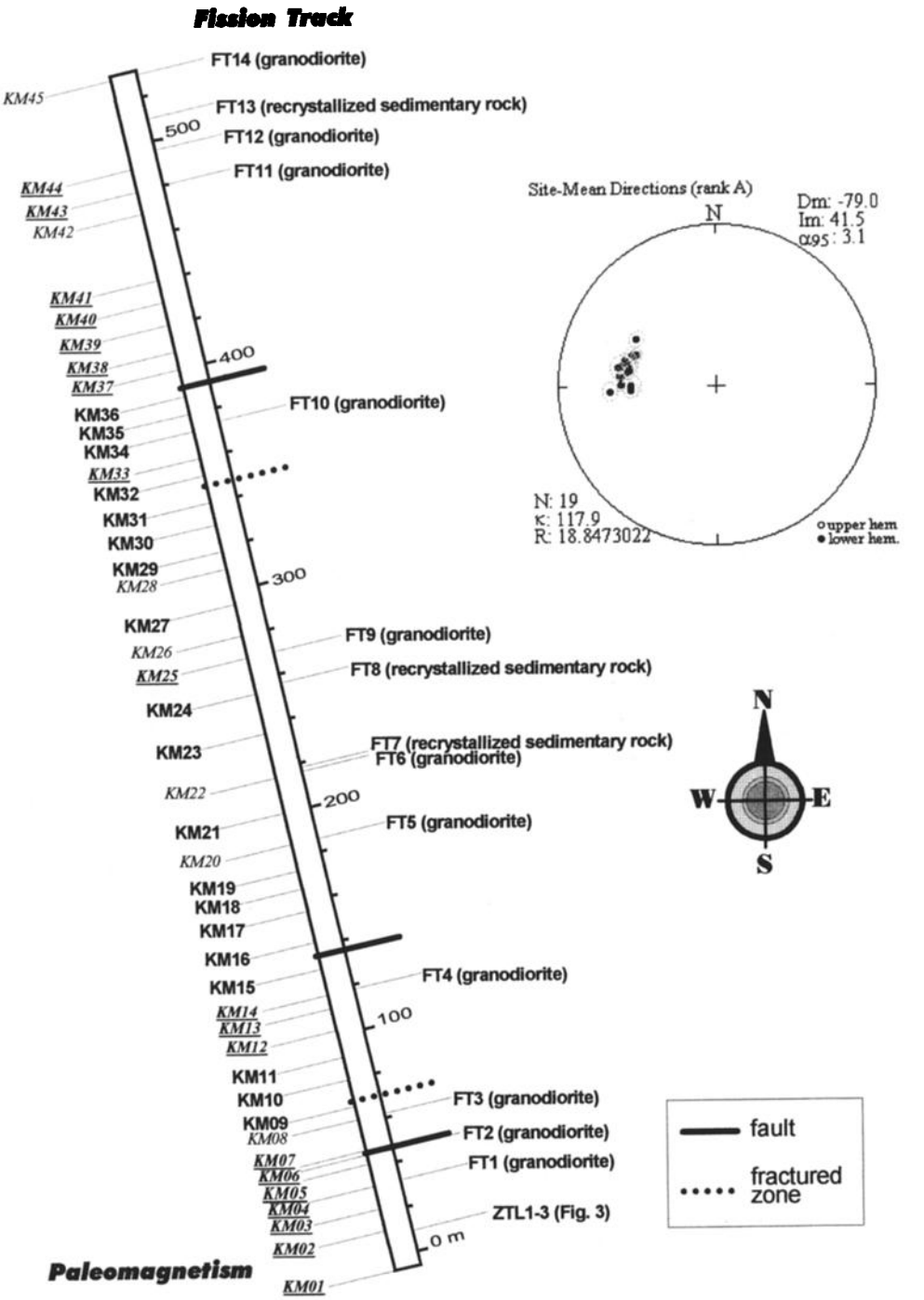


Figure 2. Sampling sites for paleomagnetic and fission track analyses in the study interval of the 250 masl drift, with the fault distribution and previous paleomagnetic results shown [*Itoh et al.*, 1998a]. See text for nomenclature of samples.

the minor intrusion of diorite; some aplite and lamprophyre veins, under map scale, also exist in the study area.

Paleomagnetic and fission track sampling was conducted along a NNW-SSE wing of the 250-m level drift in the Kamaishi Mine, away from the iron mineralization. Sampling localities are described as north-northwestward distances from a reference point (0 m in Figures 1 and 2; 39°19′50″N, 140°35′19″E). The Kurihashi Granodiorite was sampled at ~10-m intervals from 0 to the 440-m point within the drift where the granodiorite distributed with many xenoliths of diorite and hornfels. Visible E-W trending fractures were mapped at this place [*Itoh et al.*, 1998a], and secondary minerals had precipitated on the surfaces of most fractures. Larger faults (fracture zone accompanied with fault breccia) and fracture zones are also shown in Figure 2. North of the 440-m point, granodiorite and hornfels are alternately observed.

All samples were collected from the eastern wall of the drift, and the granodiorite was sampled at 45 sites for paleomagnetic analysis (KM series in Figure 2). Each paleomagnetic site contains three or four hand samples within a 3-m interval

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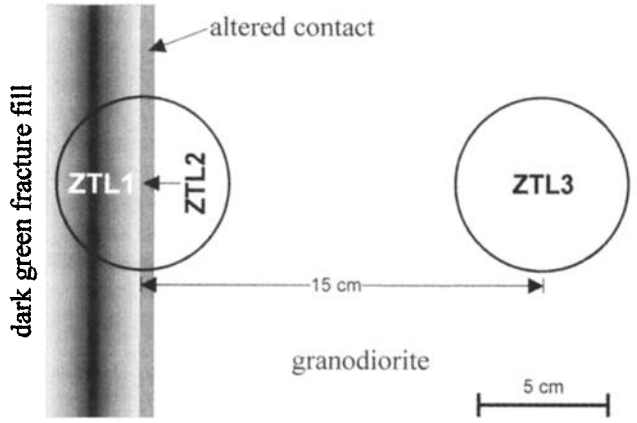


Figure 3. Schematic sampling method for fission track thermochronological analysis. Bold circles depict sidewall coring positions on the eastern wall of the 250 masl drift. See Figure 2 for locality of these samples.

which are remote from the alteration adjacent to large fractures. Each sample was independently oriented using a magnetic compass. During sampling, the magnetic north direction was verified by referring to the drift's strike to avoid any local or artificial magnetic disturbances. For conventional fission track dating, 14 block samples of ~2 kg mass were taken from the granodiorite and hornfels (FT series in Figure 2). In order to assess hydrothermal activity, samples for fission track age and length analyses were obtained from sidewall cores near the 0-m point (ZTL series in Figure 2); the sampling strategy is schematically shown in Figure 3. Fracture fill (indicative of a hydrothermal event; ZTL1), altered contact zone (ZTL2), and host granodiorite (ZTL3) were cut from the sliced core samples.

3. Paleomagnetism

Cylindrical specimens 25 mm in diameter and 22 mm in length were cut from each hand sample, after which each site contained 8–12 specimens. Magnetic susceptibility was measured for all specimens using a Bartington susceptibility meter (Figure 4). *Itoh et al.* [1998a] conducted progressive thermal demagnetization (PThD) and progressive alternating field demagnetization (PAFD) tests for selected pilot specimens from all 45 sites. They found that samples from 19 sites (italic and underlined KM site labels in Figure 2) preserved stable single-component TRM, and that seven sites (italic KM site labels) showed overlapping unblocking spectra of magnetic components or completely unstable behavior during the PThD and PAFD tests. In this study, we conducted PThD tests for the remaining 19 sites (bold-faced KM site labels in Figure 2) in an attempt to separate the remanent magnetic components. Remanent magnetizations were measured using a spinner magnetometer (Natsuhara-Giken, SMM-85), and PThD was performed, up to 640°C in air, using a noninductively wound electric furnace with an internal residual magnetic field of <10 nT. Demagnetization tests were interrupted when specimens broke into numerous pieces because of repeated heating and cooling procedures.

3.1. Remanence Direction

As shown by the PThD test, the Kurihashi Granodiorite contains a stable single-component remanence near the southern (Figures 5a–5d) and northern (Figures 5e–5h) ends of the underground sampling interval [*Itoh et al.*, 1998a]. These sites are also characterized by high Qn ratios (Figure 4). The specimens show straightforward univectorial demagnetization to the origin on orthogonal vector plots, the dominant behavior, and have unblocking temperatures (T_{UB}) between 560 and 600°C. In some sites a weak remanent magnetization with T_{UB} higher than 600°C remains throughout PThD (KM04; Figure 5b), and little change in the direction of magnetization occurs after thermal treatment at temperatures higher than the Curie temperature of magnetite (580°C). PThD result in the vicinity of a large fault (KM06; Figure 5c) suggests that the magnetization character is identical with other sites. The single-component directions are tightly clustered with large westerly deflections and are regarded as the characteristic remanent magnetization of the Kurihashi Granodiorite [*Itoh et al.*, 1998a].

In contrast, low Qn ratios (Figure 4) and multicomponent magnetizations (Figure 6) characterize the central part of the sampling interval although the natural remanent magnetization (NRM) intensity is comparable with that in the single-component sites. A high-temperature component (560°C < T_{UB} < 600°C) with westerly directions similar to the single-component direction was isolated, as was a middle-temperature component (400°C < T_{UB} < 560°C; Figures 6a–6d) with northerly or easterly declinations. In some cases, directions of the middle-temperature component differ significantly between hand samples from the same site (KM30; Figures 6b and 6c). A low-temperature component (250°C < T_{UB} < 400°C; Figures 6e–6h) is also identified in 11 sites, and its easterly deflected direction with fairly large scatter suggests that the component is not due to a recent thermoviscous remanent magnetization (TVRM) acquired during the latest Brunhes normal-polarity chron [*Dunlop*, 1995]. We determined the direction of each component using a three-dimensional least squares analysis technique [*Kirschvink*, 1980]; site-mean directions of the multicomponent remanence are listed in Table 1.

We refer to the samples with single-component and multi-component remanent magnetizations as group A and group B, respectively, in the following sections.

3.2. Magnetic Mineralogy

To identify the magnetic minerals with the Kurihashi Granodiorite, we performed thermomagnetic analysis to determine their Curie temperatures. Two samples were selected from sites KM02 and KM35, which represent samples with the sin-

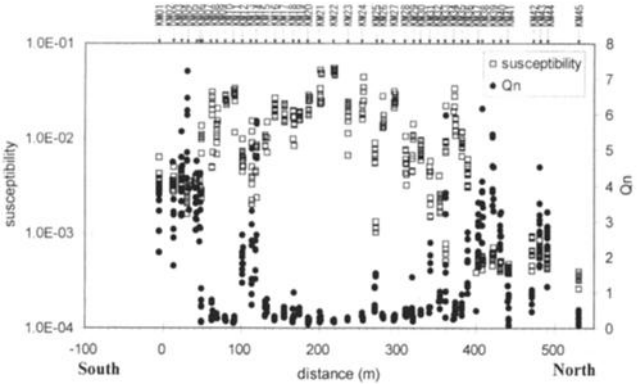


Figure 4. Magnetic susceptibility in SI units (open squares) and Königsberger ratios (Qn; solid circles) along the 250 masl drift.

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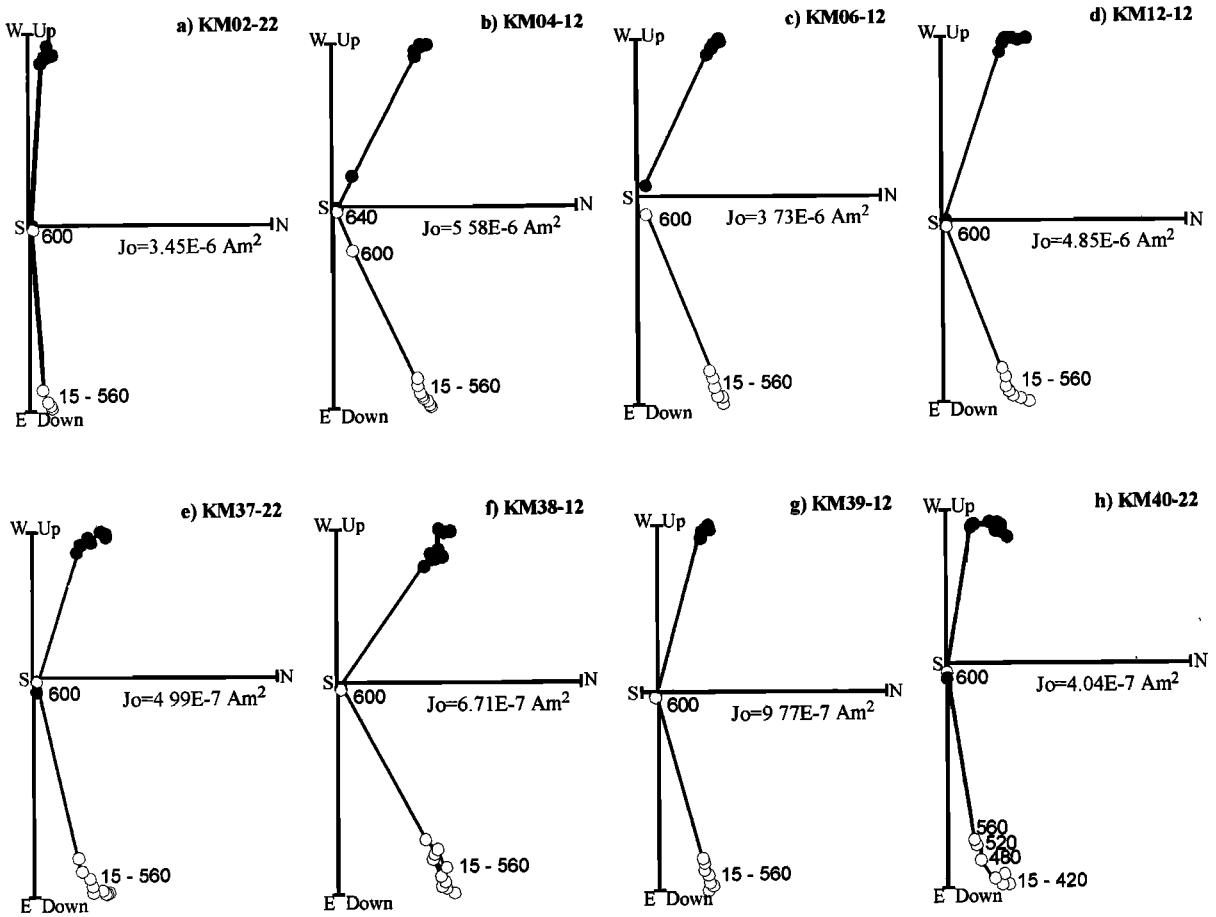


Figure 5. Examples of samples with single-component remanent magnetizations (group A) determined by the PThD test. Solid and open circles in the orthogonal projections of magnetization vector end-points are on the horizontal and N-S vertical planes, respectively. Attached numbers are temperatures in degrees Celsius.

gle-component (group A) and multicomponent (group B) remanent magnetizations, respectively (Figures 5a and 6h). Samples were prepared as small rock fragments and heated in air with an applied field of 0.8 T. The dominant magnetic mineral in both samples is Ti-poor titanomagnetite because the Curie temperatures (T_c), determined by extrapolating the upward convex portion of the heating run (Figure 7), range between 570°C and 600°C. The irreversible nature of the heating/cooling curves for sample KM02 may suggest the presence of a small amount of titanomaghemite.

We also conducted PThD of a composite isothermal remanent magnetization (IRM) induced in PAFD-treated specimens from sites KM02 and KM35. Using the procedure proposed by *Lowrie* [1990], composite IRMs were imparted by applying direct magnetic fields of 1.5, 0.4, and then 0.12 T to the samples in three orthogonal directions. Decay curves of the IRM components from PThD (Figure 8) indicate that the dominant magnetic phase is the soft coercivity fraction (<0.12 T) with a broad distribution of T_{UB} up to 580°C. Thus it is confirmed that the stable NRM of the Kurihashi Granodiorite resides in titanomagnetite as indicated by the representative samples.

3.3. Hysteresis Properties

Hysteresis parameters were determined with an alternating gradient magnetometer (Princeton Measurements Corporation, MicroMag 2900) located at Kyoto University. Again, representative samples of the two groups were selected from sites KM02 and KM35, as with the thermomagnetic analysis, and gently ground in a mortar. From the ground material, 50 rock chips as large as 1 mm were randomly selected from each site. In addition, a portion of the powder (~50 mg) was wrapped in plastic film to prepare 50 samples per site. Because the felsic minerals are more resistant to the grinding process, they tend to be concentrated in the chip samples, whereas the more fragile mafic minerals tend to concentrate in the powder samples. Values of saturation magnetization (J_s), saturation remanence (J_{rs}) and coercive force (H_c) were determined for all samples from hysteresis loops, and values of coercivity of remanence (H_{cr}) were determined from backfield demagnetization experiments.

Correlation plots of J_{rs}/J_s versus H_{cr}/H_c [*Day et al.*, 1977] are shown in Figure 9. As for site KM35, both chip and powder samples show a narrow inverse relationship on the plot, close to the trends for synthetically grown and crushed natural magnetites (trend a [*Dunlop*, 1986]) and for a mixture of single-domain (SD) and multidomain (MD) sized magnetite grains (trend b [*Channell and McCabe*, 1994]). Powder samples in KM02 follow the same trend, although some chips show unusually high H_{cr}/H_c and higher J_{rs}/J_s. Their values are plotted between trends for synthetic magnetite and for mixtures of stable SD and superparamagnetic (SP) grains (trend c [*Jackson*, 1990; *Jackson et al.*, 1993]). Thus the hysteresis properties indicate differences in grain-size distribution between the sin-

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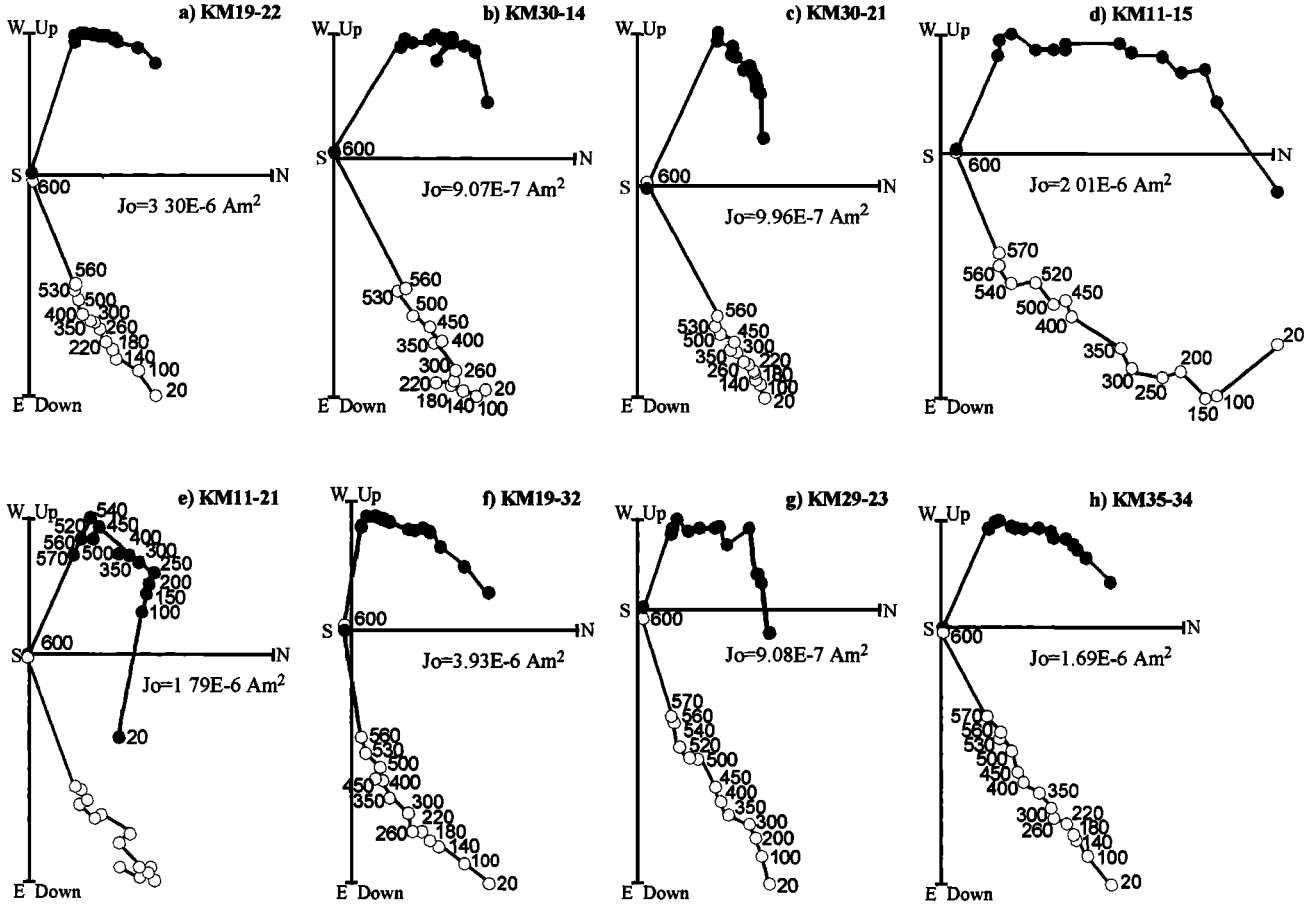


Figure 6. Examples of samples with multicomponent remanent magnetizations (group B) determined by the PThD test. Solid and open circles in the orthogonal projections of magnetization vector end-points are on the horizontal and N-S vertical planes, respectively. Attached numbers are temperatures in degrees Celsius.

gle-component and multicomponent remanence samples (groups A and B).

3.4. Frequency Dependence of Magnetic Susceptibility

As a quick check for the presence of SP grains, the frequency dependence of bulk initial susceptibility (F factor [e.g., *Heller et al.*, 1991]) was measured for thermally untreated specimens of groups A and B. Grains with sizes near the SP/SD boundary eventually become blocked and behave as SD particles when measured at higher frequency, contributing less to the overall susceptibility signal than when measured at lower frequency. Figure 10 indicates that a small but significant frequency dependence exists in group A specimens. Thus the presence of finer magnetite grains in the single-component remanence

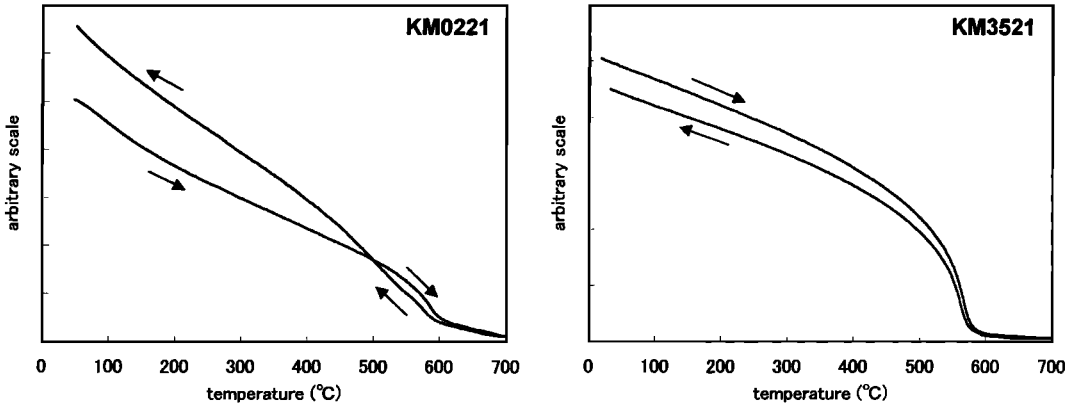


Figure 7. Thermomagnetic curves of representative samples of the Kurihashi Granodiorite representing samples with single-component (KM02) and multicomponent (KM35) remanent magnetization. Measurements were done on small rock fragments in air with an applied field of 0.8 T.

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samples is supported, although a quantitative estimate is difficult from the F factor which is a function of the grain-size distribution of the ferrimagnetic mineral [*Eyre*, 1997].

3.5. Microscopic Observation

On the basis of the results of the PAFD test, values of the median destructive field (MDF) are compared for the two groups of samples (Figure 11). MDF values of many group B samples are considerably lower, suggesting the dominance of MD magnetite grains. This result is consistent with microscopic observations, which show 10–100 μm magnetites in the group B sample (KM3521 in Figure 12). In sharp contrast, MD magnetite is very rare in the group A sample (KM0213 in Figure 12). Hence we conclude that SD magnetite grains and a mixture of MD and SD magnetite grains carry the remanent

Table 1. Site-Mean Paleomagnetic Directions of the Kurihashi Granodiorite

Site	Dm	Im	α_{95}	k	N
High-Temperature Component (T_{UB} = 560–580°C)					
KM09	−56.6	57.3	5.5	121.1	7
KM10	−64.2	52.5	5.7	114.0	7
KM11	−64.7	47.3	4.8	158.7	7
KM15	−57.7	41.8	9.5	41.7	7
KM16	−45.5	−3.0	5.4	153.9	6
KM18	−74.4	39.0	3.6	284.1	7
KM19	−78.5	42.6	3.6	285.3	7
KM23	−56.4	49.6	5.9	104.7	7
KM27	−46.2	55.5	7.3	69.8	7
KM29	−58.1	48.3	6.4	91.3	7
KM30	−58.5	43.8	3.9	241.4	7
KM31	−71.6	45.8	4.8	158.6	7
KM32	−79.6	45.4	4.4	186.9	7
KM34	−68.9	43.7	5.4	126.5	7
KM35	−68.8	45.5	5.0	144.0	7
KM36	−69.3	43.8	3.1	374.6	7
Middle-Temperature Component (T_{UB} = 400–560°C)					
KM09	−2.7	38.8	19.4	16.6	5
KM10	47.2	48.4	13.5	21.1	7
KM11	20.0	41.5	14.4	18.4	7
KM15	18.3	51.2	14.9	17.5	7
KM16	13.2	21.7	12.6	23.8	7
KM18	6.5	39.4	16.3	14.6	7
KM19	4.5	47.4	12.0	26.1	7
KM23	36.5	32.2	16.5	14.4	7
KM27	34.8	54.7	9.1	45.1	7
KM29	16.2	53.8	10.6	33.2	7
KM30	18.9	53.6	14.0	19.5	7
KM31	−5.8	42.3	21.1	9.1	7
KM32	10.2	38.3	18.9	13.5	6
KM34	25.9	40.0	15.9	15.4	7
KM35	2.4	52.6	7.3	69.2	7
KM36	0.8	51.5	17.6	12.8	7
Low-Temperature Component (T_{UB} = 250–400°C)					
KM09	23.8	24.1	20.9	20.3	4
KM11	63.5	18.5	29.1	6.3	6
KM15	68.4	30.1	19.4	10.7	7
KM16	50.6	−14.6	16.5	14.3	7
KM18	47.4	17.6	21.4	13.7	5
KM19	45.0	29.1	9.3	52.3	6
KM27	63.2	26.3	17.3	20.5	5
KM29	72.1	20.9	10.1	44.5	6
KM30	45.6	36.5	20.1	21.8	4
KM35	31.8	33.3	16.3	14.6	7
KM36	37.3	34.4	20.8	14.5	5

Dm and Im, site-mean declination and inclination in degrees; α_{95}, radius of 95% confidence circle; k, *Fisher*'s [1953] precision parameter; N, number of specimens; T_{UB}, unblocking temperature.

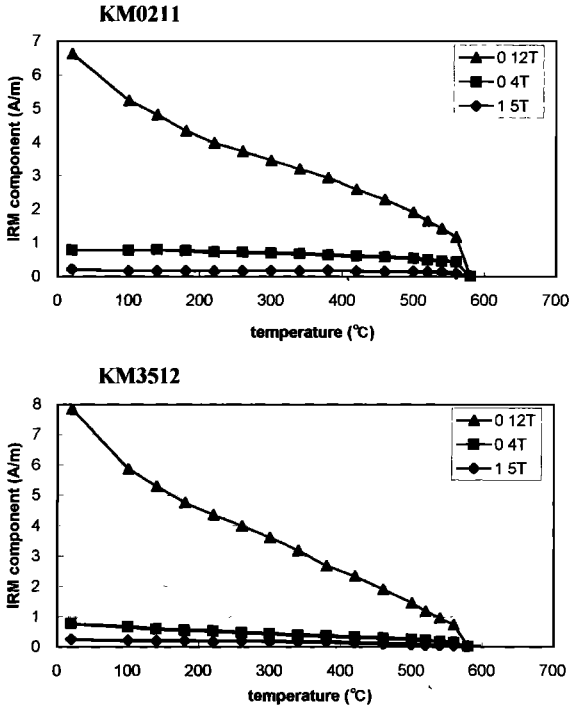


Figure 8. Thermal demagnetization curves of orthogonal IRMs on samples with single-component (KM02) and multicomponent (KM35) remanent magnetizations.

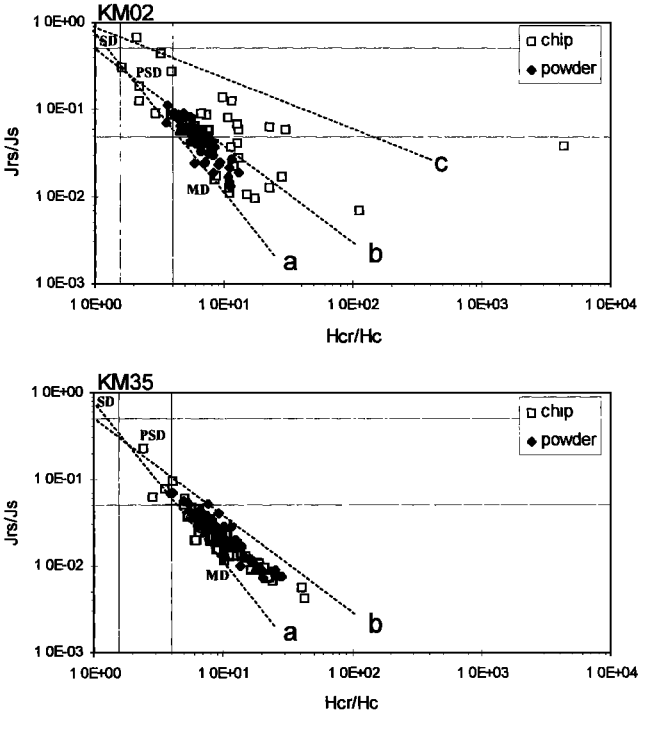


Figure 9. Logarithmic plots of hysteresis parameters representing samples with single (KM02) and multicomponent (KM35) remanent magnetizations. See text for preparation method of chip and powder samples. Trends a, b, and c originate from synthetic and crushed natural magnetite [*Dunlop*, 1986], single-domain (SD) and multidomain (MD) mixture of magnetite [*Channell and McCabe*, 1994] and superparamagnetic (SP) and SD mixture of magnetite [*Jackson*, 1990; *Jackson et al.*, 1993], respectively.

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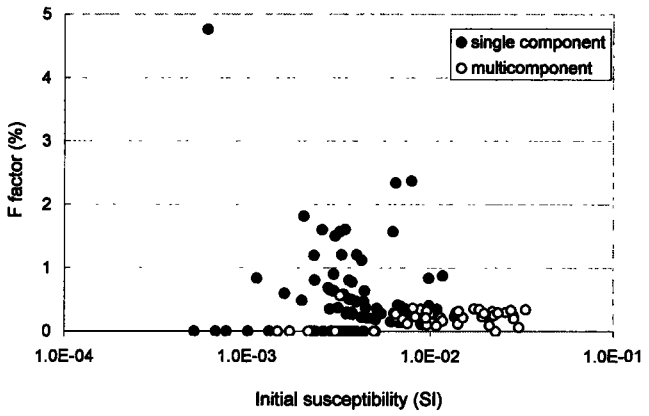


Figure 10. Bulk initial susceptibility measured at 0.47 kHz versus the F factor ($100 \times (\chi_{0.47kHz} - \chi_{4.7kHz})/\chi_{0.47kHz}$ [*Heller et al.*, 1991]) for specimens with single (solid circle) and multicomponent (open circle) remanent magnetizations.

magnetizations of group A and group B of the Kurihashi Granodiorite, respectively.

4. Fission Track Analysis

4.1. Initial Cooling Age of Kurihashi Granodiorite

We obtained zircon grains sufficient for fission track dating from samples of 10 sites within the Kurihashi Granodiorite. Dates were obtained using the external detector method on the internal surface of zircon grains (ED1 [*Gleadow*, 1981]) to avoid the external effect of plutonic rocks caused by intergranular uranium. Samples were etched in KOH-NaOH eutectic etchant (1:1) at 225°C for 22 hours. Neutron irradiation of samples was conducted in the TRIGA MARK II nuclear reactor at St. Paul's University (Rikkyo Daigaku), Japan. Ages were calculated using the ζ calibration method [*Hurford*, 1990], with a ζ factor of 370 ± 4 for NBS-SRM612 dosimeter glass [*Danhara et al.*, 1991]. Table 2 lists results of the fission track dating. The 10 sites (FT1–FT14) yield ages that are statistically identical with a mean age of 98.3 ± 5.3 Ma.

4.2. Evidence of Reheating by Hydrothermal Activity

Fission track ages and length distribution analyses were conducted in order to assess the effect of hydrothermal activity. Three samples were prepared exclusively from fracture fill, altered contact, and fresh granodiorite (Figure 3). Fortunately, all of them contain enough zircon grains for fission track analyses. Mineralogical properties of the zircons in the fracture fill are similar to those of zircons in the granodiorite, suggesting that they originated from the host rock. Fission track dating of these samples was undertaken following the routine procedure outlined in section 4.1. As shown in Table 2 and Figure 13, zircons from fracture fill (ZTL1) and the altered contact (ZTL2) yield mean ages significantly younger than the granodiorite (ZTL3) age which is identical with those of other granodiorite samples (FT series in Table 2) constituting a "plateau" age of the sampled interval (Figure 2).

For the three samples (ZTL1-3), fission track lengths and azimuths in the zircon crystals were determined using a microscopic three-dimensional measurement system [*Yamashita et al.*, 1992; *Iwano et al.*, 1996]. Measurements were conducted for confined tracks revealed both as tracks in tracks (TINTs) and as tracks in cleavages (TINCLEs). Within the fracture fill (ZTL1 in Figure 13), track lengths show a unimodal distribution with an average length of 10.9 μm. This length is comparable with the reference value of 10.5 ± 0.1 μm determined for zircon age standards from rapidly cooled volcanics [*Hasebe et al.*, 1994]. In contrast, the track length distribution for the granodiorite sample (ZTL3) is characterized by a "tail" of shortened tracks <9 μm, although the average length does not differ from the reference value. The histogram for zircon track lengths in the altered contact (ZTL2) has an intermediate appearance.

Combining age and length variations, we suggest that an annealing event of fission tracks by hydrothermal activity postdated emplacement and initial cooling of the plutonic rock. In fractures filled with high-temperature liquid, preexisting tracks would have been completely annealed under elevated temperatures, and a mean age calculated for subsequent unannealed tracks would provide the timing of cessation of hydrothermal circulation. Moving away from the liquid's path, its thermal impact would diminish as suggested by older mean ages. The existence of shorter fission tracks indicates that the samples were heated up to the partial annealing zone of zircon (200–300°C on a geologic timescale [*Yamada et al.*, 1995]). The "plateau" age of the FT series Kurihashi Granodiorite samples and ZTL3, with obvious signs of annealing in track length

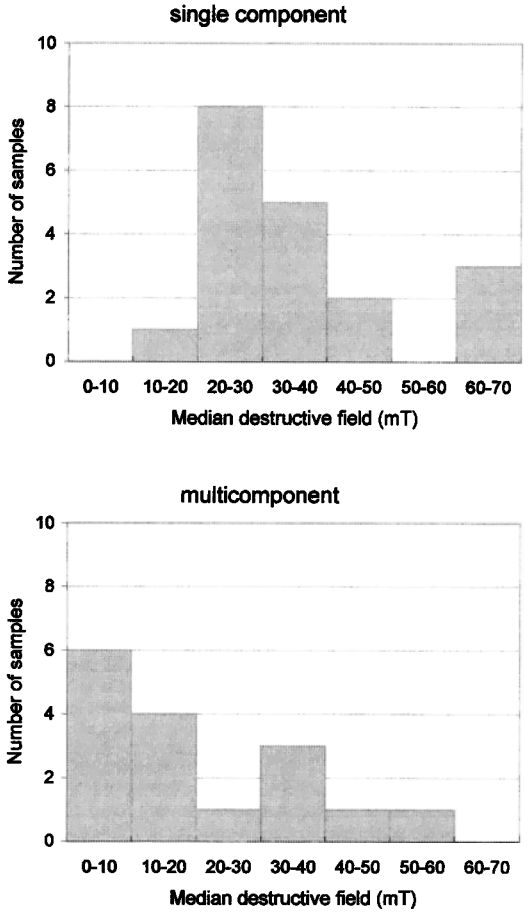


Figure 11. Histograms of median destructive fields during the PAFD test for sites in which (top) single- component and (bottom) multicomponent remanent magnetizations were determined by the PThD test.

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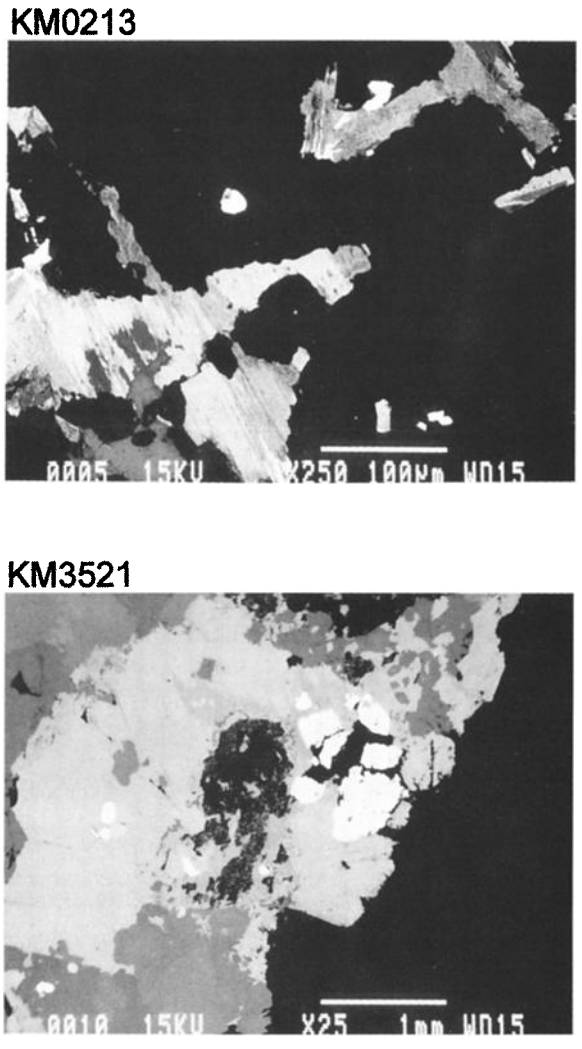


Figure 12. Examples of backscattered electron images of opaque minerals in the Kurihashi Granodiorite from representative samples with single-component (KM02) and multicomponent (KM35) remanent magnetizations. Bright spots in the KM0213 and KM3521 images are hematite and magnetite, respectively. Scale bars are shown at the bottom of the images.

distribution, may be affected by the postcooling thermal event as well.

A significant difference in thermal history between the fracture fill and host granodiorite is supported by the analysis of fluid inclusions along healed microcracks in quartz grains. On the basis of *Roedder*'s [1984] methods, *Amano et al.* [1998] analyzed samples taken from a shallower drift of the Kamaishi Mine (300 m above the present study level; see Figure 1b for projected locality). They showed that homogenization temperatures in the fracture fill cover the range from 170°C to 310°C, while those in the Kurihashi Granodiorite (5 cm from alteration zone) range from 130°C to 210°C with minor exception (Figure 14). Thus the hydrothermal circulation seems to have been active in many parts of the plutonic body which is covered by dense fracture network.

5. Discussion

5.1. Origin of Multicomponent Remanent Magnetization

Fission track thermochronology has indicated that the Kurihashi Granodiorite underwent a postcooling thermal event. On the basis of the natural long-term annealing characteristics of fission tracks in zircon [*Yamada et al.*, 1995; *Tagami and Shimada*, 1996], paleotemperatures during this thermal event reached ~230–330°C for a heating duration of $\sim 10^6$ years at the sampling point of the Kurihashi Granodiorite. Figure 15 shows two sets of thermoviscous remagnetization contours: those representing the SD blocking model of *Pullaiah et al.* [1975] (thick solid curves) and those representing the SD relaxation model of *Middleton and Schmidt* [1982] (thin solid curves). The horizontal arrow shows the range of maximum T_{UB} for the "middle-temperature" component, indicated by relatively sharp junctions of linear vector components on the PThD orthogonal plots for the group B samples. In the SD time-temperature relation, the T_{UB} range in the laboratory corresponds to the remagnetization temperatures in nature (shown as shaded oval). However, many of the group B samples obtained from the Kurihashi Granodiorite have anomalously high T_{UB}, some of which are within 20°C of T_c of magnetite and are incompatible with either set of contours.

As stated in section 3.5, group B samples contain large truly

Table 2. Results of Zircon Fission Track Dating for the Kurihashi Granodiorite in the Kamaishi Mine

Site	Number of Grains	Spontaneous Track ρ_s, cm^{-2}	Spontaneous Track N_s	Induced Track ρ_i, cm^{-2}	Induced Track N_i	$P(\chi^2)$, %	Dosimeter ρ_d, cm^{-2}	Dosimeter N_d	r	U Content, ppm	Age ± 1σ, Ma
FT1	30	1.02×10^7	4989	1.57×10^6	772	17	8.20×10^4	1264	0.262	150	97.3 ± 4.8
FT3	30	9.51×10^6	5454	1.58×10^6	905	67	8.31×10^4	1280	0.554	150	92.0 ± 4.3
FT4	30	1.03×10^7	5506	1.51×10^6	809	43	8.21×10^4	1265	0.422	150	103.0 ± 5.0
FT5	30	8.85×10^6	4856	1.23×10^6	675	88	8.32×10^4	1281	0.691	120	110.0 ± 6.0
FT6	30	9.87×10^6	4555	1.55×10^6	715	35	8.22×10^4	1266	0.526	150	96.2 ± 4.8
FT9	30	9.67×10^6	5549	1.54×10^6	884	24	8.23×10^4	1267	0.644	150	94.9 ± 4.5
FT10	30	9.55×10^6	6888	1.40×10^6	1010	96	8.23×10^4	1268	0.800	140	103.0 ± 5.0
FT11	30	8.48×10^6	8162	1.32×10^6	1268	80	8.24×10^4	1269	0.793	130	97.4 ± 4.1
FT12	30	8.07×10^6	6561	1.34×10^6	1087	0	8.25×10^4	1270	0.273	130	91.5 ± 4.1
FT14	30	8.01×10^6	8099	1.24×10^6	1251	3	8.25×10^4	1271	0.550	120	98.0 ± 4.2
Samples for Thermochronological Analysis											
ZTL1	30	9.42×10^6	3911	1.79×10^6	743	50	8.63×10^4	1329	0.473	170	83.5 ± 4.1
ZTL2	30	9.87×10^6	3179	1.77×10^6	571	100	8.60×10^4	1325	0.636	170	88.0 ± 4.8
ZTL3	30	1.02×10^7	5356	1.64×10^6	862	24	8.58×10^4	1321	0.619	150	97.9 ± 4.6

Here ρ and N are density and total number of counted tracks, respectively. $P(\chi^2)$ is probability of obtaining the χ^2 value for ν degrees of freedom (ν is number of grains minus 1); r is the correlation coefficient between ρ_s and ρ_i.

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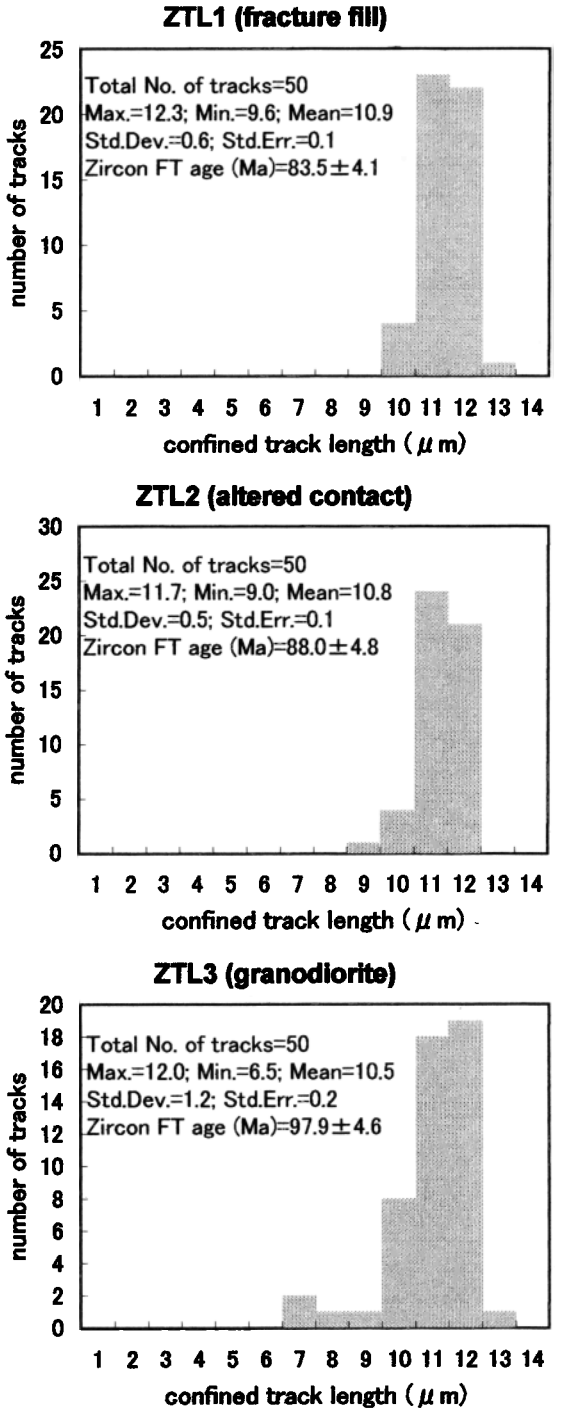


Figure 13. Ages and confined fission-track lengths in zircon samples obtained from fracture fill (ZTL1), altered contact rock (ZTL2), and granodiorite (ZTL3). See Figure 3 for sampling scheme.

MD grains of Ti-poor magnetite. Because the partial TRM and VRM carried by MD magnetite have long thermal demagnetization tails [*Xu and Dunlop*, 1994; *McClelland et al.*, 1996], the elevated maximum T_{UB} range in the Kurihashi Granodiorite could be due to MD magnetite. On the other hand, TRMs of the group A containing SD magnetite as the dominant ferrimagnetic mineral were not affected by the thermal event because T_{UB} spectra of TRM of SD grains are concentrated below T_c in most cases [*Dunlop and Özdemir*, 1997].

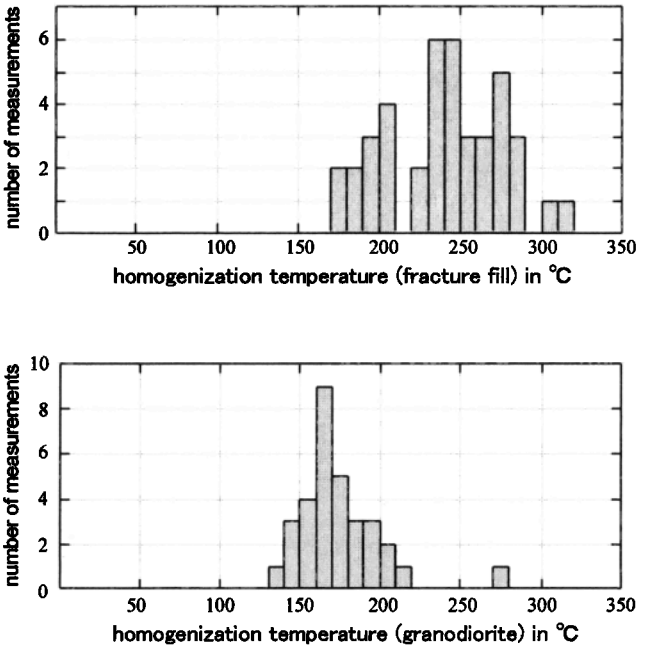


Figure 14. Histograms of homogenization temperatures of fluid inclusions in quartz from fracture fill and granodiorite. See Figure 1b for projected sampling locality.

It is noteworthy that the annealed fission track ages in the fracture fill and altered contact (Figure 13) can be correlated with the Geomagnetic Superchron in the Late Cretaceous [*Gradstein et al.*, 1995]. Long-term reheating under a constantly normal geomagnetic field resulted in considerable overprinting of the thermoviscous remanent magnetization (TVRM) of the Kurihashi Granodiorite.

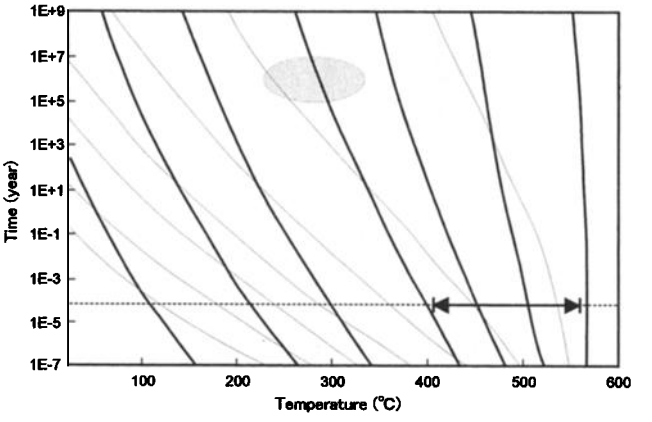


Figure 15. Thermoviscous remagnetization contours predicted by *Pullaiah et al.* [1975] on the basis of *Néel*'s [1949] SD blocking theory (thick solid curves) and by *Middleton and Schmidt* [1982] on the basis of the SD relaxation theory of *Walton* [1980] (thin solid curves). Shaded oval depicts plausible remagnetization temperatures reached in nature, estimated from the partial annealing zone of zircons for a heating duration of $\sim 10^6$ years [*Tagami and Shimada*, 1996]. Horizontal arrow shows the range of laboratory unblocking temperatures of the "middle-temperature component" residing in the Kurihashi Granodiorite.

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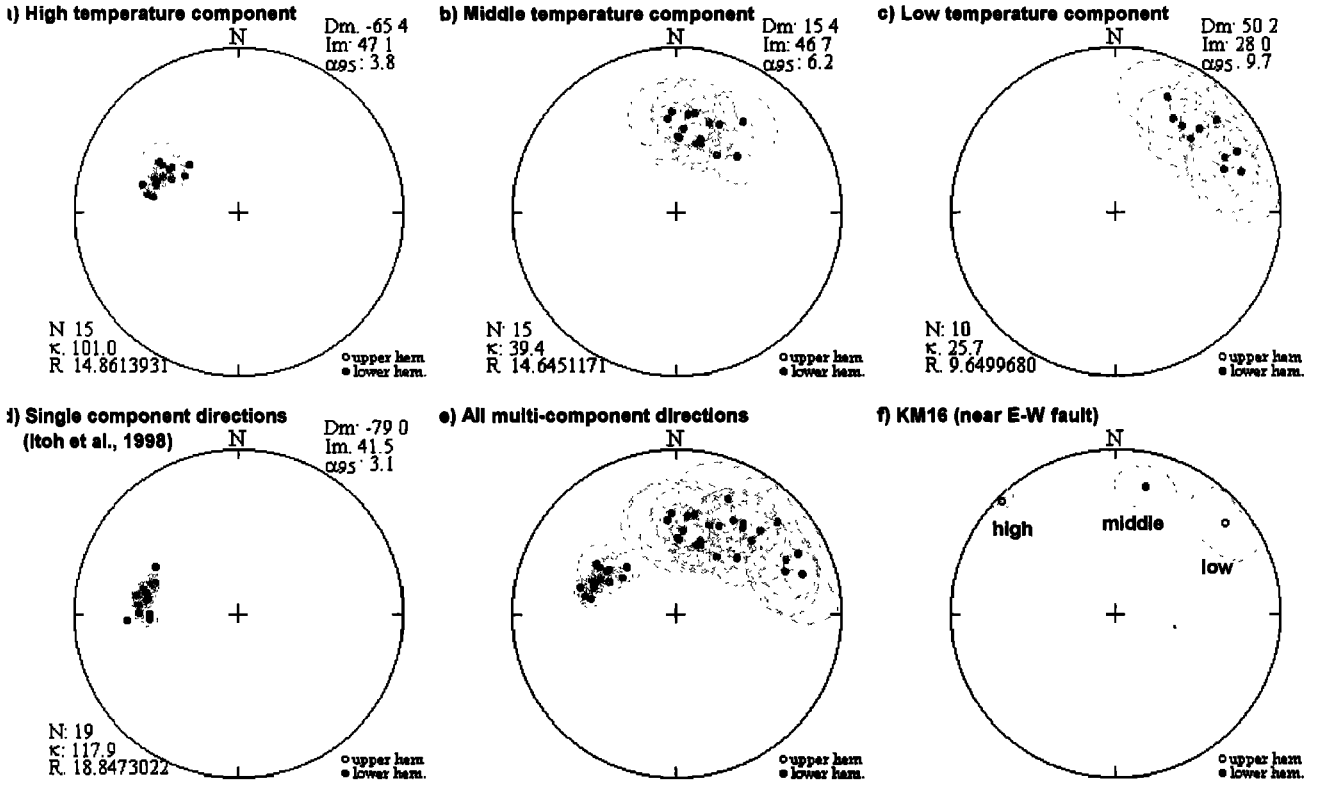


Figure 16. Equal-area plots of site-mean magnetic directions with circles of 95% confidence obtained from the Kurihashi Granodiorite. Solid and open symbols are on the lower and upper hemispheres, respectively.

5.2. Cretaceous Counterclockwise Rotation in Northeast Japan

From our paleomagnetic and thermochronological analyses we have recognized and characterized high-, middle-, and low-temperature remanence in the Kurihashi Granodiorite. Site-mean directions of the high-temperature component (Figure 16a) are similar to those of the stable single-component TRM (Figure 16d) reported by *Itoh et al.* [1998a]. Middle- and low-temperature components of remanence (Figures 16b and 16c) have northerly and easterly site-mean directions, respectively. Because the individual components have been found in the same rock specimens, these directions can not be explained by local differential motion within the sampled interval of the 250 m above sea level drift. We propose that the difference in direction of the three components is due to a counterclockwise rotation of the Kurihashi Granodiorite in northeast Japan during the Late Cretaceous. A comparison of the mean directions of the westerly and easterly components indicates a rotation angle (R) and its uncertainty (ΔR) defined by *Beck* [1980] of $-116° \pm 12°$.

Passing from the high- to low-temperature components, the scatter in site-mean directions increases, and the low-temperature directions have an elongate distribution on the equal-area net (Figure 16c). The larger scatter may be brought about by differences in hydrothermal activity due to the density of open fractures in the granodiorite and the duration of fluid circulation between site or even within a site. As exemplified by discordant middle-temperature directions from the same site (Figures 6b and 6c), heterogeneous hydrothermal activity may have resulted in differential timing of remanence acquisition and recording of rotational phases. Figure 16e shows an equal-area plot of all site-mean directions obtained in this study. It appears that the directions are aligned in a west to east trend, implying that TVRM acquisition was contemporaneous with the Late Cretaceous counterclockwise rotation.

For site KM16, the site-mean directions of the three components show systematically shallower inclinations than the other sites (Figure 16f). Since this site is adjacent to a large fault (Figure 2), we suspect that the directions were affected by local drag along the fault and have omitted them from the calculation of characteristic mean directions.

5.3. Tectonic Implications of the Rotational Events

Figure 17 summarizes the paleomagnetism and thermochronology of the Kurihashi Granodiorite. Radiometric dates with various closing temperatures [*Yoshida et al.*, 2000] indicate rapid initial cooling of the plutonic body. Rapid uplift and exhumation of the Cretaceous plutonic rocks in northeast Japan was also reported by *Itoh et al.* [1998b]. They inferred from geomagnetic anomaly modeling, borehole stratigraphy and paleomagnetic measurements that the acoustic basement of the forearc shelf of northeast Japan was composed of the Early Cretaceous granites unconformably overlain by the Albian marine sediments. Thus the stable TRMs of the granites in northeast Japan, which have consistently westerly deflections, seem to have been acquired during the course of the regional uplift in the late Early Cretaceous. Uplift and cooling also resulted in fracture formation influenced by intragranular microcracking of quartz grains [*Vollbrecht et al.*, 1991], which acted as the path of hydrothermal liquid in subsequent stages.

After initial uplift and cooling, the Kurihashi Granodiorite underwent hydrothermal alteration as indicated by the differ-

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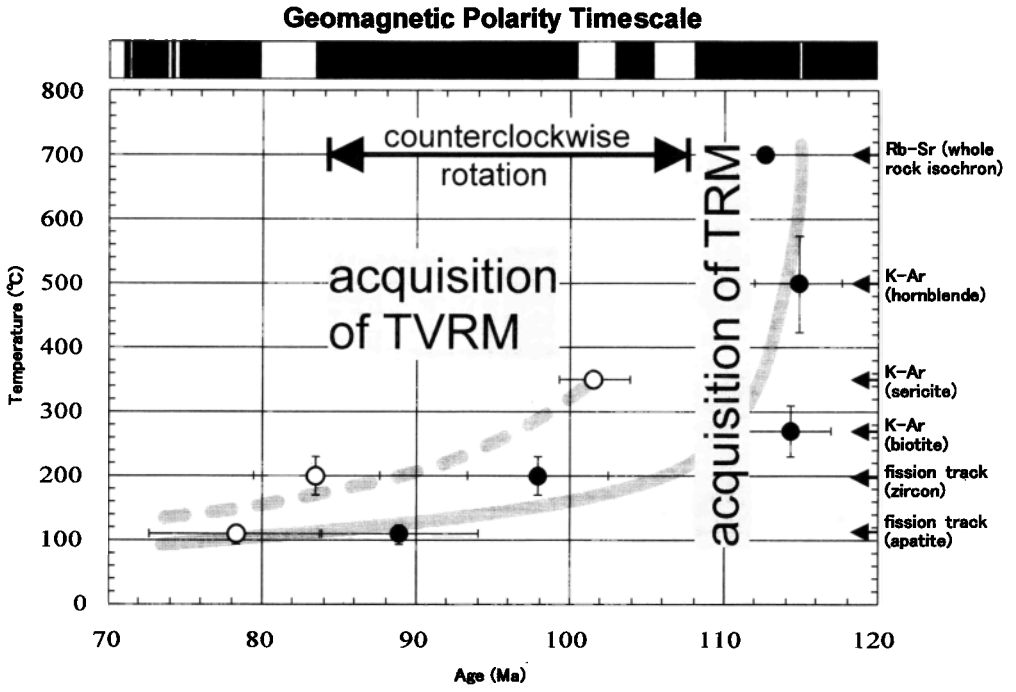


Figure 17. A summary of the cooling history and rotational motion of the Kurihashi Granodiorite. Radiometric dates obtained from the host granodiorite (solid circle) and fracture fill (open circle) are after *Yoshida et al.* [2000]. On the basis of cooling paths (solid and dashed bold lines) we estimate the timing for the acquisition of primary TRM and thermoviscous remanent magnetization (TVRM). Standard geomagnetic timescale is after *Gradstein et al.* [1995].

ence in cooling paths between the granodiorite and fracture fill during the Late Cretaceous (Figure 17). We believe that the TVRMs described in the previous sections were acquired during this period of elevated temperatures and continuously normal geomagnetic field. Presence of two TVRM components in 11 sites and large scatter in directions of the low-temperature component can be attributed to intermittent circulation that is inferred from fracture-filling mineral assemblages corresponding to different temperatures [*Kano et al.*, 1999]. Although the effect of tectonic tilting can not be corrected for in granites, sedimentary rocks of the late Early Cretaceous Miyako Group and Late Cretaceous Kuji Group along the Pacific coast are unconformably settled on granitic rocks without large tilting [*Oide et al.*, 1989]. Therefore the present result indicates that a part of northeast Japan rotated counterclockwise in the Late Cretaceous and then rotated clockwise. This result is quite discordant from previous tectonic models [e.g., *Otofuji et al.*, 1985b, 1994] which regard northeast Japan as a coherent block rotated counterclockwise during the Miocene formation of the Japan Sea.

Otsuki [1992] has reviewed geologic and tectonic data for the Japanese Island and has argued that the eastern margin of Eurasia was a site of left-lateral faulting in the Cretaceous according to highly oblique north-northwestward subduction of the Izanagi plate. In his model the transcurrent faulting ceased at 85 Ma because the subduction direction changed from oblique to normal. On the basis of age we assume that a block containing the Kurihashi Granodiorite rotated counterclockwise under left-lateral active faulting along the Cretaceous Eurasian margin. As for the clockwise rotation suggested by the easterly deflection of the low-temperature component, we have no constraint on timing or tectonic implication. As determined by *Otofuji et al.* [1985b], paleomagnetic data near large faults in northeast Japan tend to have "discordant" easterly deflections. Recently, *Yamaji et al.* [1999] studied Miocene rocks near the Nihonkoku-Miomote Tectonic Line, and found easterly deflected paleomagnetic directions from a zone (Figure 1a) much wider than previously estimated [*Otofuji et al.*, 1985b]. They argued that northeast Japan had been rifted as several rigid blocks bounded by right-lateral wrench zones during the Miocene formation of the Japan Sea under a transtensional stress regime. If this is the case, a simple tectonic model of an intact northeast Japan being rifted and rotated by a fan-shaped opening of the Japan Sea [*Otofuji et al.*, 1994] should be reconsidered in light of smaller-scale tectonic movements.

6. Conclusions

1. Paleomagnetic data for the Upper Cretaceous Kurihashi Granodiorite can be divided into two groups: Group A shows a stable single component of TRM, and the group B has two or three distinct components of remanence. Components with high, middle, and low unblocking temperatures (T_{UB}) have westerly, northerly, and easterly deflected declinations, respectively.

2. The predominant magnetic mineral in the Kurihashi Granodiorite is magnetite with $T_c \cong 580°C$.

3. SP magnetite grains exist in the group A samples based on bulk hysteresis measurements. A frequency dependence of the initial susceptibility in the same samples also supports the presence of SP grains.

4. The magnetite in the group B samples, detected by bulk hysteresis measurements, is of SD to MD size. Low MDF

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values in the same samples also suggest a considerable amount of MD magnetite, and microscopic observations confirm the MD size ($\geq$100 μm) of the magnetite grains.

5. Fission track (FT) thermochronology using zircon indicates that the Kurihashi Granodiorite underwent a hydrothermal event. Differences in the thermal effects between the host granodiorite and authigenic fracture fill imply that hydrothermal circulation was active in the fracture network of the plutonic body. Mean FT ages suggest that the event ceased in the Late Cretaceous.

6. T_{UB} of the overprint NRM in the group B samples are too high to be described by thermoviscous remagnetization theories considering the maximum temperatures estimated by FT thermochronology. Anomalous T_{UB} may be explained by long thermal demagnetization tails in MD magnetite.

7. The Kurihashi Granodiorite, initially intruded in the Early Cretaceous, was rapidly uplifted and underwent a post-cooling thermal event. The difference in declinations among the multicomponent NRMs in the granodiorite might record a counterclockwise rotation in northeast Japan. Considering the Cretaceous stress regime along the eastern margin of Eurasia, the rotational event could be related to left-lateral wrench deformation.

8. Easterly declinations of the low-temperature component require a clockwise rotation since the Cretaceous; it is discordant with a hypothesis of coherent tectonic block rotation in northeast Japan during Cenozoic back arc opening and is suggestive of differential motion of smaller blocks.

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